

DHRUV PATEL

Computer Science Student | Software Engineering & AI/ML | NJIT

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PROFESSIONAL SUMMARY

Computer Science student at NJIT (GPA 3.689) who builds software, generative AI, and machine learning projects in Python, including a retrieval-augmented generation app, a fraud detection model, and a secure banking API, and works as a Level 3 Process Assistant at Amazon. Seeking internships in software engineering and AI where I can build reliable, well-tested systems.

EDUCATION

New Jersey Institute of Technology (NJIT)

Newark, NJ

B.S. in Computer Science | **GPA: 3.689 / 4.0** | *Student Excellence Award*

Expected May 2028

Coursework: Databases, Operating Systems, Discrete Mathematics, Data Science, Compiler Design, Information Systems

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, SQL, PHP, HTML, CSS
AI & ML	LLM APIs, RAG, prompt engineering, embeddings, FAISS, scikit-learn, pandas, NumPy
Software Engineering	FastAPI, Flask, REST APIs, Docker, Git & GitHub, GitHub Actions (CI/CD), pytest, JWT auth
Databases & Systems	PostgreSQL, MySQL, SQLAlchemy, database design, Vista, SCADA, yard management
CS Concepts	Data structures & algorithms, OOP, operating systems, discrete math, root-cause analysis

PROFESSIONAL EXPERIENCE

Process Assistant (L3) – Inbound, Mezz & Outbound Operations

Amazon Sortation Center – Carteret, NJ

Aug 2024 – Present

- Operate and troubleshoot Amazon's internal process software (Vista, Dock Console, OB SSP, FMC) across inbound, Mezz, and outbound, resolving system errors fast to keep package records accurate and dispatch on schedule during every shift.
- Monitor SCADA-controlled conveyor and sortation systems and yard and dock tools, escalating faults to maintenance and IT teams to limit downtime, protect throughput, and keep trailers departing on schedule for every scheduled CPT departure.
- Trace failed moves and data discrepancies in Vista, reconciling system records with physical inventory to catch defects early, coach the associate involved, and keep them from growing into larger inventory or customer delivery problems.
- Redesigned a consolidation workflow, cutting a 20-pallet load to 11 pallets (about 45% fewer) and reducing ad hoc transportation, and train and coach associates on procedures, safety, and system usage while covering for Area Managers.

ACADEMIC PROJECTS

AI Document Q&A Assistant (RAG) | Python, FastAPI, FAISS, LLM API

Sep 2026

- Built an end-to-end retrieval-augmented generation app: uploaded PDF, text, and Markdown files are split into overlapping chunks, embedded with sentence-transformers, and stored in a persistent FAISS cosine-similarity index that reloads on restart.
- Answers questions by retrieving the top matching passages and passing only those to an LLM API (Claude) to write a cited answer, with an offline fallback that returns the best passage when no API key is set, so the app always keeps working.
- Exposed upload, ask, and health endpoints through FastAPI with input validation and clear error handling, and wrote 7 pytest tests covering chunking, embeddings, retrieval quality, disk persistence, and the API, plus clear setup documentation.

Secure Banking Transactions API | FastAPI, PostgreSQL, Docker, GitHub Actions

Sep 2026

- Designed a banking REST API in FastAPI and SQLAlchemy with bcrypt password hashing, short-lived JWT access tokens, and role-based access, so customers can only reach their own accounts while admins can view all accounts and users.
- Stored money as integer cents and ran each transfer in a single database transaction with row locks taken in a fixed order, preventing lost updates, overdrafts, and deadlocks between simultaneous transfers, so account balances always stay correct.
- Containerized the API with Docker Compose and PostgreSQL, wrote 9 pytest tests covering authentication, permissions, and transfer rules, and set up a GitHub Actions pipeline that runs the tests on every push to keep the main branch stable.

Fraud Detection ML Model | Python, pandas, scikit-learn

Sep 2026

- Trained and compared Logistic Regression, Random Forest, and Gradient Boosting on the Kaggle Credit Card Fraud dataset (284,807 transactions, only 0.173% fraud), using stratified splits that keep the same fraud share in every subset.
- Handled the severe class imbalance with class and sample weights, tuned each model's decision threshold on validation data, and judged results with precision, recall, F1, ROC-AUC, and PR-AUC instead of accuracy, which is misleading on this data.
- Selected the Random Forest as the best model (0.864 precision, 0.829 recall, 0.842 PR-AUC on held-out data) and added prediction and reporting scripts with 5 pytest tests and a written analysis of trade-offs between the three models.

LEADERSHIP & INVOLVEMENT

President – Gujarati Undergraduate Student Association (GUSA), NJIT: leads board and 200+ person events

Jan 2025 – Present

Event Coordinator – Hindu YUVA Club, NJIT: plans cultural and educational events with volunteer teams

Sep 2024 – Present